

Roll No. \_\_\_\_\_

TU/CSE

TEZPUR UNIVERSITY

Mid Semester Test, Autumn 2018

C0 504 Natural Language Processing (Alternative)

Marks : 40

Time : 75 minutes

The figures in the right-hand margin indicate marks for the individual question.

1. Write Perl regular expressions to recognise the strings  
(i) “What”, “what”, (ii) positive integers? 2
2. Make the correct pairs from terms in list A with those in list B- 4  
A: (A1)Unicode (A2)UTF-16 (A3)“Times New Roman”  
(A4)ISCII (A5)Gurmukhi  
B: (B1)Font (B2)ISO-10646 (B3)Script (B4)Variable length encoding (B5)8-bit encoding
3. \_\_\_\_ (Huffman / Nyquist) Theorem gives an \_\_\_\_ (upper / lower) limit of *sampling* frequency for \_\_\_\_ (analog-to-digital / digital-to-analog) conversion of a signal? 3
4. (a) In n-gram analysis *smoothing* is done to remove errors in the training corpus. True or False? 1  
(b) Probability of a bi-gram “a b” is estimated as *count of occurrences* of bi-gram “a b” divided by \_\_\_\_ (total number of bi-grams in the corpus / count of occurrence of “a” / count of occurrence of “b”). 1  
(c) Name two smoothing methods in the context of n-grams. 2
5. (a) Name three Parts-of-Speech tagging approaches? 3  
(b) What are tags possible for the words- *that, can, talk, fast, have, question*? 3

(c) Consider the following text-

The reporters know that he can talk very fast.  
They have come ready with their questions  
and recording devices.

- (i) Suggest the sequence of tags appropriate for the words in the text given above. (Give a dash if you are not sure about a particular word.) 3
- (ii) How do we know the correct PoS for “fast” in the above sentence. 1
6. Give an example of *left recursion* in a context free grammar? 1
7. (a) Give an example each of noun phrases, verb phrase and prepositional phrase. 3  
(b) Give examples of a declarative and an imperative sentence in English and suggest a context free grammar for such sentences. 2 + 2
8. (a) Point out the phenomenon of *agreement* in an English sentence. 2  
(b) Give examples to show the phenomenon of *sub-categorisation* of verbs. 2
9. What kind of ambiguity is there in the following sentence? 1  
I saw him waiting for the bus.
10. (a) Earley’s parser is a \_\_\_\_ (top-down / bottom-up) parser. 1  
(b) Name two actions in Earley’s parser? 2  
(c) What would be the output of Earley’s parser for an ambiguous sentence like the one in question 9? 1